CONFIDENTIAL

MID TERM TEST II (PRACTICAL)
SEMESTER I, 2022/2023

SUBJECT CODE	**:**	**SECJ1023**
SUBJECT NAME	**:**	**PROGRAMMING TECHNIQUE II**
SECTION	**:**	**1 - 11, and 15**
TIME / DURATION	**:**	**8PM – 10PM / 2 HOURS**
DATE / DAY	**:**	**16 MAY 2023 / TUESDAY**
VENUE	**:**	**COMPUTER LABS**

INSTRUCTIONS

- This is a CLOSE-BOOK practical test. You will attempt the examination by doing coding on a computer.
- You are provided with some reference resources. You are only allowed to refer to the provided reources.
- Refeences to other resources including any source code, any websites, etc. is strictly prohibited.
- Using any form of artificial intelligence tool is strictly prohibited. Such tools include ChatGPT, AI-based VS Code extensions like Co-Pilot, etc.
- Do coding offline using any C++ IDE such as VS Code, DevCPP, etc.
- You are allowed to use your own laptop.
- You will be using eLearning only for downloading the provided resources and submitting the answer.Please refer to your class eLearning for the submission.
- All COMMENTS in the program WILL NOT BE GRADED.
- Programs that CANNOT COMPILE will get 50% PENALTY.
- Be warned that your program will undergo a screening process for PLAGIARISM DETECTION.
- There will be NO TECHNICAL ASSISTANCE provided during the test. This include issues like IDE is not working, program cannot compile, etc.Should you face any issue,you will need to deal with it on your own.
- Any question related to the test question will not be entertained at all. Read the question carefully.

DOWNLOAD THE <u>PROVIDED RESOURCES</u> FROM ELEARNING

This question paper consists of FOUR (4) printed pages excluding this page.

There is <u>No starter code</u> provided for this test. You write the program from scratch. Write the program using inline style and use only a single source code file.

Question **[100 Marks]**

Define a class named **`WordNumber`** that models a number in English word.

The requirements for the class are as follows:

- The number is stored in English word for example, `zero`, `one`, `two`, and so on.
- The number is stored in an attribute of type string named **`data`**.
- You should consider only numbers zero to nine. Other numbers are beyond the scope of this test.
- The numbers should be treated as case-insensitive. That means, numbers `Two`, `two`, and `TWO` are equivalent.

Write a C++ program following Object-Oriented principles to accomplish the following tasks.

1. Declare the class with its attribute

(5 marks)

2. Define a constructor that accepts a parameter of type string to set the class's attribute, `data`. Declare this constructor such that it can also serve as a default constructor (or default argument constructor). In this case, the parameter should be defaulted to string of `"zero"`.

(6 marks)

3. Define another constructor that accepts an integer number as a parameter. This constructor will translate the number into an English word to be stored in the attribute, `data`. The translation should be done based on Table 1.Note that only numbers 0 to 9 is supported, other numbers will be translated to an empty string.

Table 1:The number in integer values and its English word

Int number	**word**
0	zero
1	one

2	two
3	three
4	four
5	five
6	six
7	seven
8	eight
9	nine

(11 marks)

4. Define a mutator method that sets the attribute `data` with a value of type string.

(7 marks)

5. Define a conversion method that converts a `WordNumber` object to an integer value. This method uses the string number stored in the attribute of the object to do the conversion following the translation table from Table 1. Consider only numbes zero to nine. As for other numbers, they will be converted to 0 value.

(6 marks)

6. Define an overloaded operator that will be used to determine whether a `WordNumber` object is greater than another `WordNumber` object. This operator will result in a boolean value. The comparison should be done with the integer value of the objects. You should make use of the converstion method from Task 5 to implement this operator.

(6 marks)

7. Define an overloaded operator for the addition operation to add a `WordNumber` with an integer value. This operator should results in an integer value. The addition operation is done by addding the integer value of the object with the integer number. For example, assume a `WordNumber` object, n is set to “nine”, thus the following statements:

   `n + 11` will result in `20`.

(5 marks)

8. Define another overloaded operator for the addition operation to add a `WordNumber` with an string and results in a string. The addition operation is done by concatenating the the object's `data` with the string. For example, assume a `WordNumber` object, `n` is set to "`nine`", thus the following statements:
   `n + string(" thousands")` will result in "`nine thousands`".

(5 marks)

9. Define another overloaded operator for the addition operation to add a `WordNumber` object with another `WordNumbr` object This operator will results in another `WordNumber` object. The addition operation is done by adding the integer values of both objects and translate the result into an English Word to be stored in the attribute `data`. For example, consider two `WordNumber` objects, `a` and `b` are set to "`one`", and "`two`", respectively. Adding these two objects will result in another object with its attribute `data="three"`.

(6 marks)

10. Define a regular functions named **`printObject`** and specify it as a friend to the class `WordNumber`. This function accepts a `WordNumber` object as parameter amd prints out the `data` of the object along with its integer value. For example **`(seven, 7)`**.

(4 marks)

11. Define another regular functions named **`printArray`** to print a list of `WordNumber` objects that are passed as parameters. This function should make use of function from Task 10.

(6 marks)

12. In the main function, declare an array to hold 4 elements of `WordNumber` objects. Initialize each element with any value using the constructor as follows:
    - First element with the default constuctor
    - Second and third elements with the constructor from Task 2.
    - Fourth element with the constructor from Task 3.

    Then, print all the objects of the list.

(10 marks)

13. Modify the data of each object in the list using input from the user and print the new list

(6 marks)

14. Using a loop and appropriate methods and operators of the class, find the largest object and the sum of the objects of the list. Then print the largest and sum objects.

(13 marks)

15. Using an appropriate addition operator of the class, add the sum object from Task 14 with a string value and store the result into a variable. The string value shuld be obtained from a user input.

(4 marks)

Figure 2 shows an example run of the program. The **bold text** indicate user inputs.

```
Original objects
(zero,0)
(one,1)
(two,2)
(three,3)

To modify the objects with user inputs
Enter a number in word=> two

Enter a number in word=> Zero

Enter a number in word=> TWO

Enter a number in word=> thrEE

Modified objects
(two,2)
(zero,0)
(two,2)
(three,3)

The sum:  (seven,7)

The largest: (three,3)

To add object with a string
Enter a string=> Wonders
The result of addding object with the input string
sevenWonders
```

Figure 2

-Emd Of Question--